

QUIZ

Write down the formula for calculating the mean of continuous random variable X .

INDEPENDENT RANDOM VARIABLES

- We saw that:

If X, Y independent random variables $E(XY) = E(X)E(Y)$

- Now variance under independence:

If X, Y independent then $Var(X + Y) = Var(X) + Var(Y)$

Proof:

INDEPENDENT RANDOM VARIABLES

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If X, Y independent then $Var(X + Y) = Var(X) + Var(Y)$

Proof:

$$Var(X + Y) = Var(X) + Var(Y) + 2Cov(X, Y)$$

$$Cov(X, Y) = E(XY) - \mu_X \mu_Y = \mu_X \mu_Y - \mu_X \mu_Y = 0$$

INDEPENDENT RANDOM VARIABLES

- Result of the previous proof:

If X, Y independent then $Cov(X, Y) = \rho(X, Y) = 0$

- Is the converse also always true? NO!

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- Is the converse also always true? NO!

Counter Example: $X \in \{-1, 0, 1\}$ each equally likely.

Let $Y = X^2$. $Cov(X, Y) = E(XY) - E(X)E(Y) = E(X^3) - 0$

X^3 same variable as $X \Rightarrow E(X^3) = 0 \Rightarrow Cov(X, Y) = 0$

X, Y not correlated but dependent (Y determined by X).

WIDELY USED DISCRETE DISTRIBUTIONS: BERNOULLI

X has Bernoulli distribution with parameter p , if

- Only outcomes are $X = 0$ or $X = 1$ and
- $P(X = 1) = p \quad P(X = 0) = 1 - p$

Denoted with $X \sim \text{Bernoulli}(p)$

WIDELY USED DISCRETE DISTRIBUTIONS: BERNOULLI

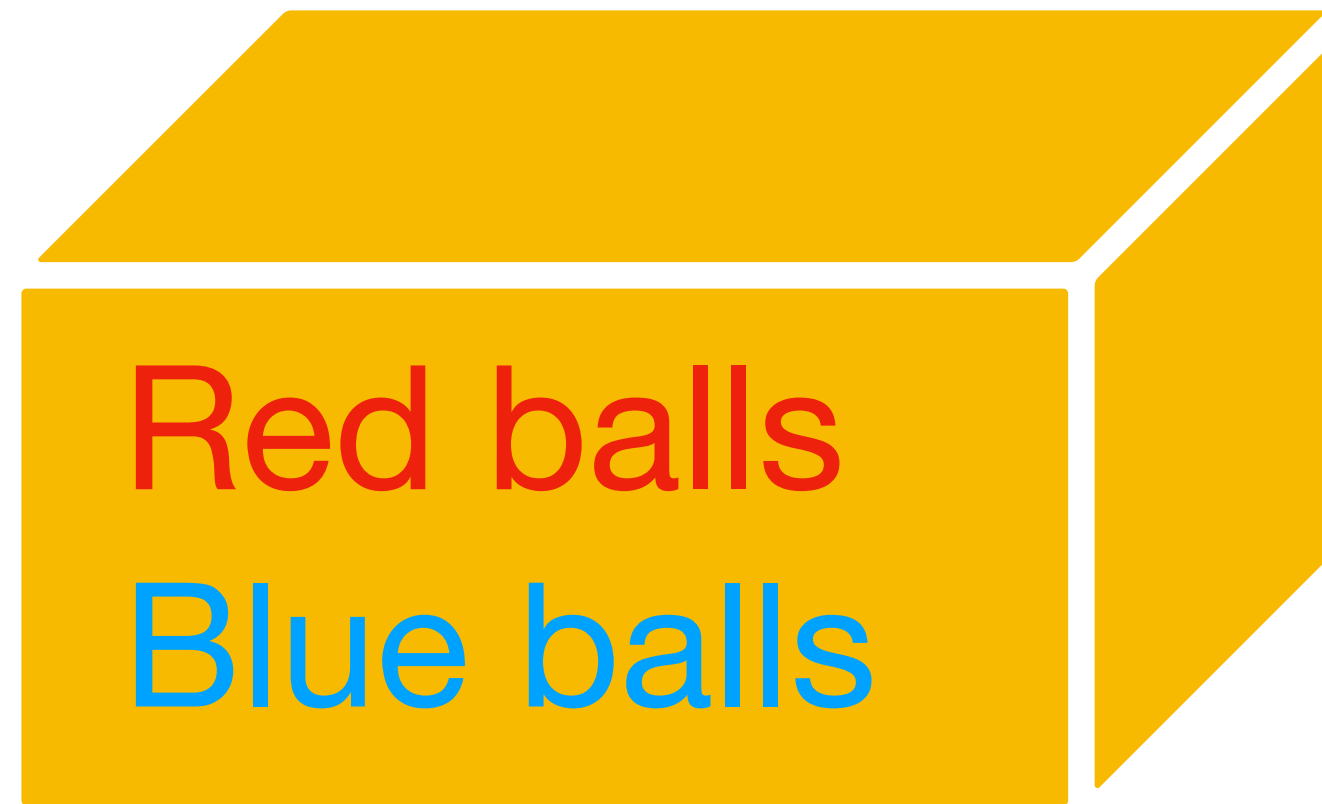
- Examples:



Tossing a coin ($X = 1$ if Head, $X = 0$ if Tail and $p = 1/2$ if fair)

WIDELY USED DISCRETE DISTRIBUTIONS: BERNOLLI

- Examples:



Drawing one R ball where proportion of R balls in the box is p

($X = 1$ if R , $X = 0$ if B and parameter p)

WIDELY USED DISCRETE DISTRIBUTIONS: BERNOULLI

- Examples:



Clinical trial: Treatment success

($X = 1$ if Successful, $X = 0$ if Fails and parameter p)

WIDELY USED DISCRETE DISTRIBUTIONS: BERNOULLI

- Expectation and Variance of Bernoulli distribution:

$$E(X) = 1 \times p + 0 \times (1 - p) = p$$

Since X and X^2 are the same random variable: $E(X^2) = E(X)$

$$Var(X) = E(X^2) - (E(X))^2 = p - p^2 = p(1 - p)$$

WIDELY USED DISCRETE DISTRIBUTIONS: BERNOULLI

- Bernoulli Trial:

If $X_1, X_2, \dots$ are iid (independently and identically distributed) and each X_i has Bernoulli distribution with parameter p
 $\Rightarrow X_1, X_2, \dots$ are Bernoulli trials with parameter p .

WIDELY USED DISCRETE DISTRIBUTIONS: BINOMIAL

X has Binomial distribution with parameters n, p , if

$$f(x) = \begin{cases} \binom{n}{x} p^x (1-p)^{n-x} & \text{for } x = 0, 1, 2, \dots, n \\ 0 & \text{otherwise} \end{cases}$$

Denoted with $X \sim \text{Binomial}(n, p)$

or $X \sim B_{n,p}$

WIDELY USED DISCRETE DISTRIBUTIONS: BINOMIAL

- Example:



A coin with probability of Head p . Toss it n times. Tosses are iid.

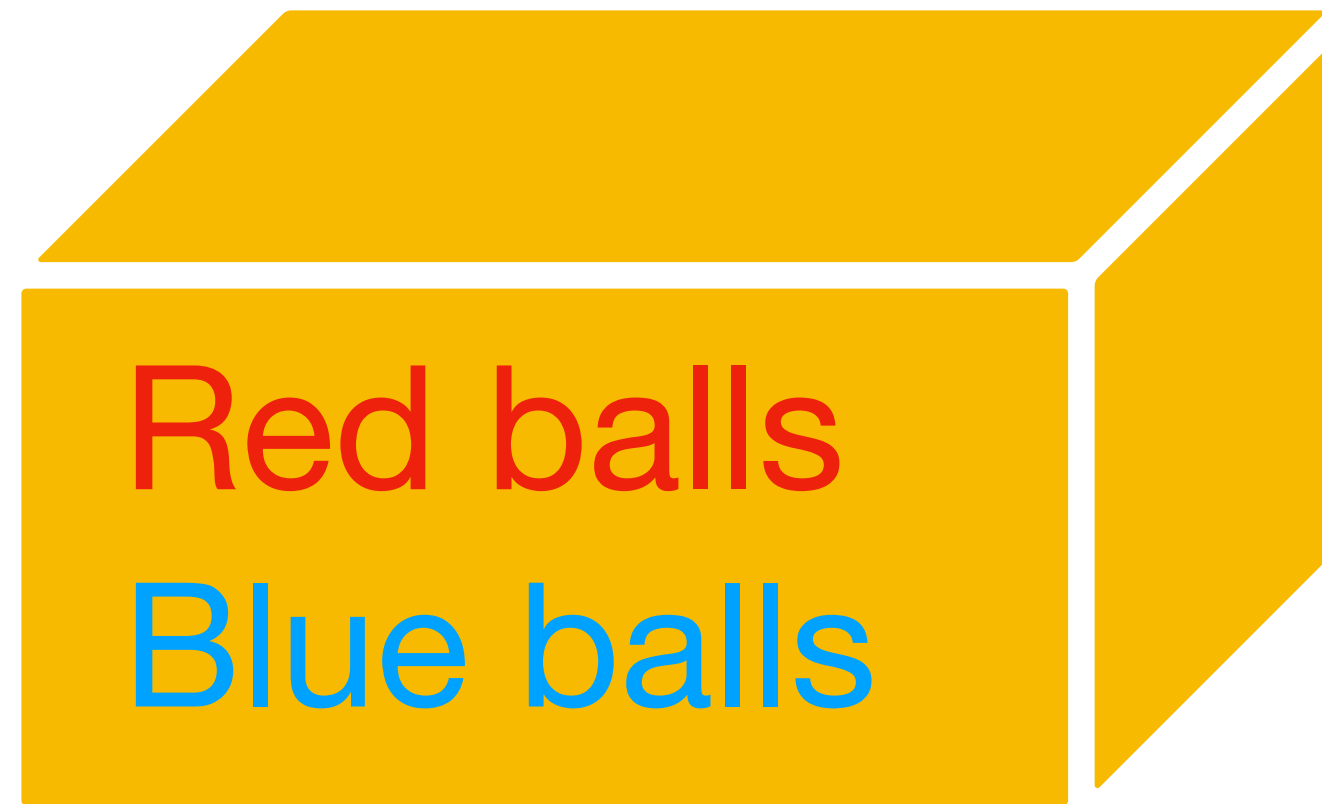
X : # of Heads in n tosses. $X \sim \text{Binomial}(n, p)$

For a particular sequence of x Heads and $(n - x)$ Tails:

probability is $p^x(1 - p)^{n-x}$. $\binom{n}{x}$ ways of choosing a sequence.

WIDELY USED DISCRETE DISTRIBUTIONS: BINOMIAL

- Example:



Proportion of R balls in a box containing R , B balls is p .

X : # of R balls in n draws with replacement.

$X_i = 1$, if i^{th} draw a R ball, $X_i = 0$ otherwise.

$$X = X_1 + X_2 + \cdots X_n, \quad X \sim \text{Binomial}(n, p)$$

WIDELY USED DISCRETE DISTRIBUTIONS: BINOMIAL

- Can generalize the previous example:

Theorem

Let $X_1, X_2, \dots, X_n$ form n Bernoulli trials and $X = X_1 + \dots + X_n$

$$X \sim \textit{Binomial}(n, p)$$

WIDELY USED DISCRETE DISTRIBUTIONS: BINOMIAL

- Expectation and Variance of Binomial distribution:

$$E(X) = \sum_{i=1}^n E(X_i) = \sum_{i=1}^n p = np$$

$$Var(X) = \sum_{i=1}^n Var(X_i) = \sum_{i=1}^n p(1 - p) = np(1 - p)$$

WIDELY USED DISCRETE DISTRIBUTIONS: BINOMIAL

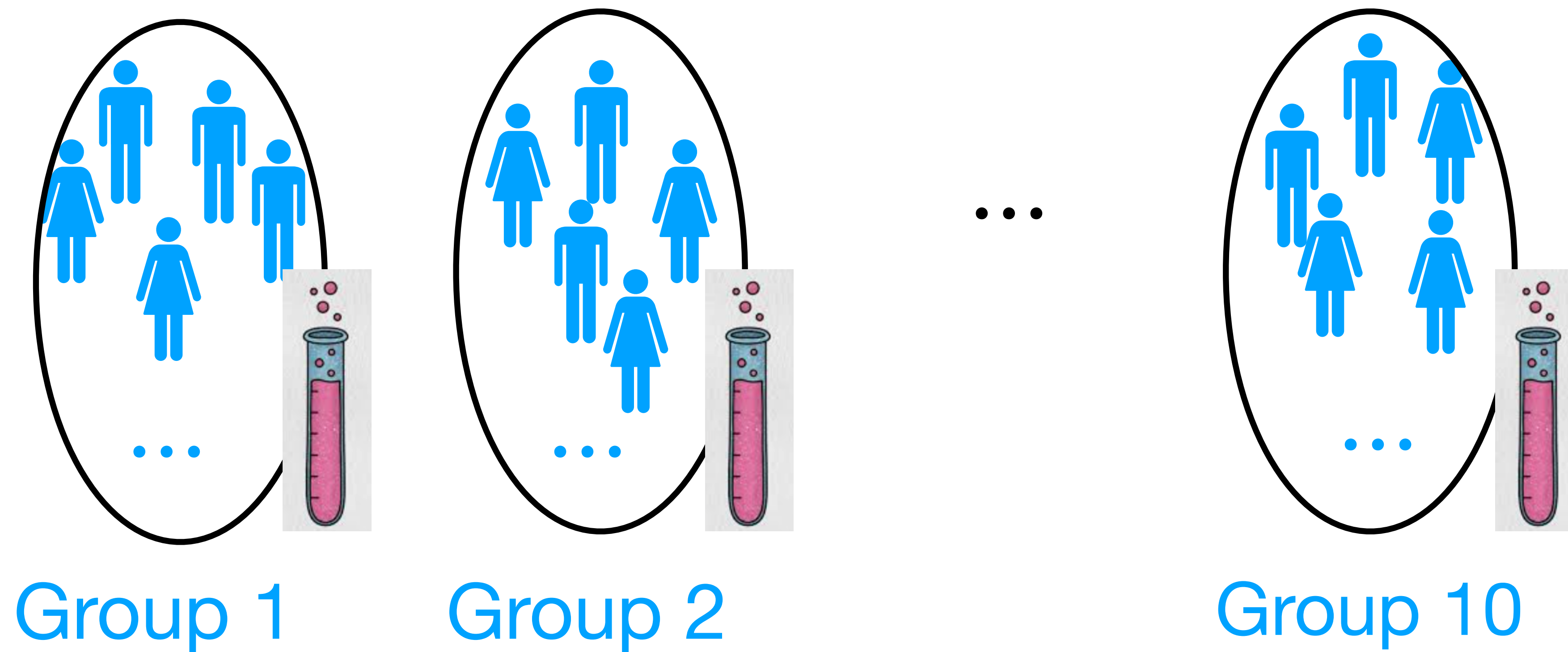
Ex: Test large group (1000) for a rare disease (affects 2 in 1000).

Naive testing requires 1000 tests.

WIDELY USED DISCRETE DISTRIBUTIONS: BINOMIAL

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Naive testing requires 1000 tests. **A better way:**

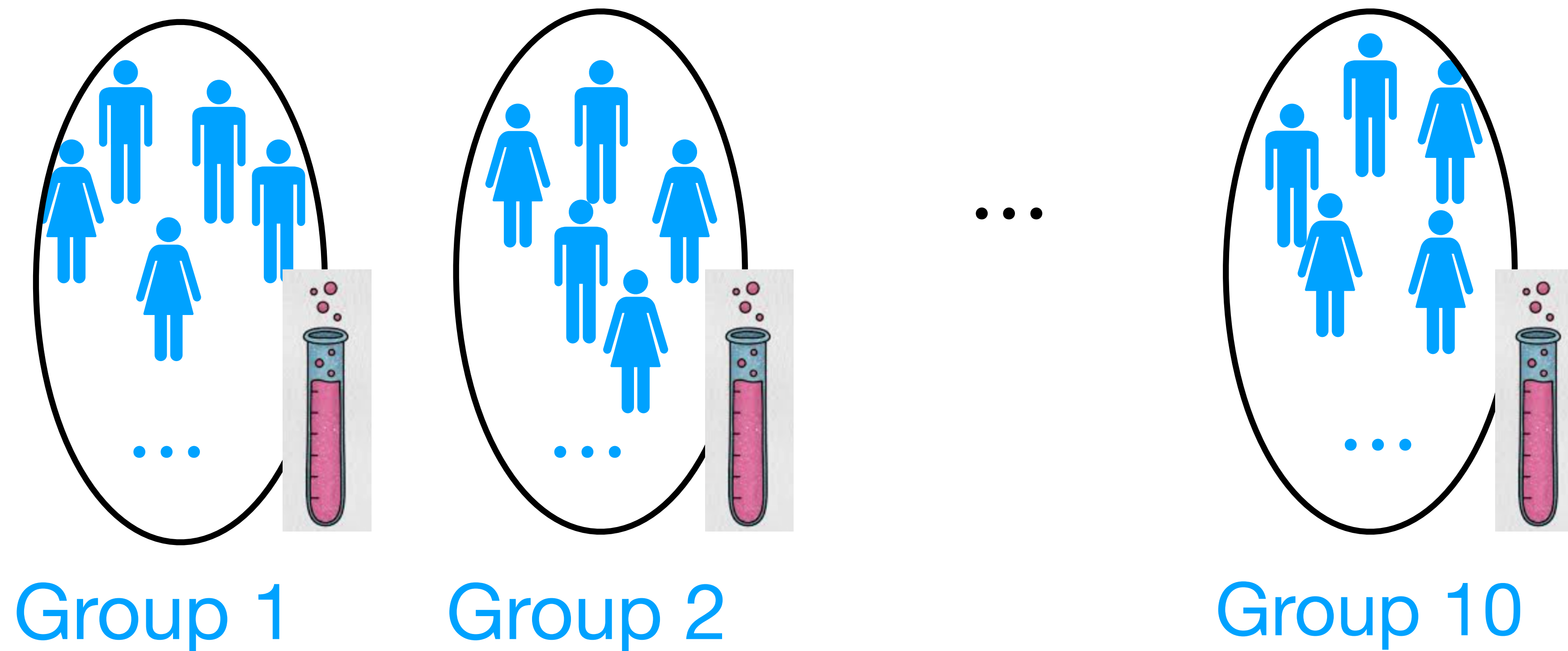


100 people in each group. Single mixed tube for each group.

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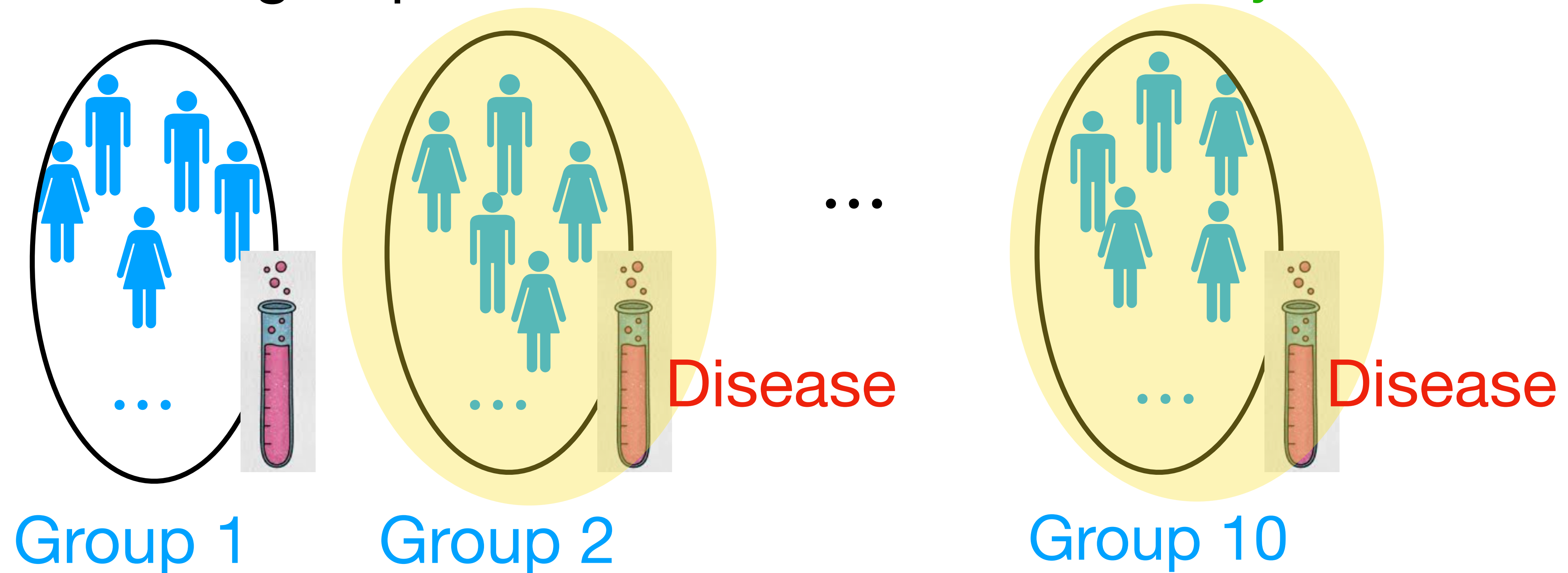


10 tests so far. If no group test shows disease we are done!

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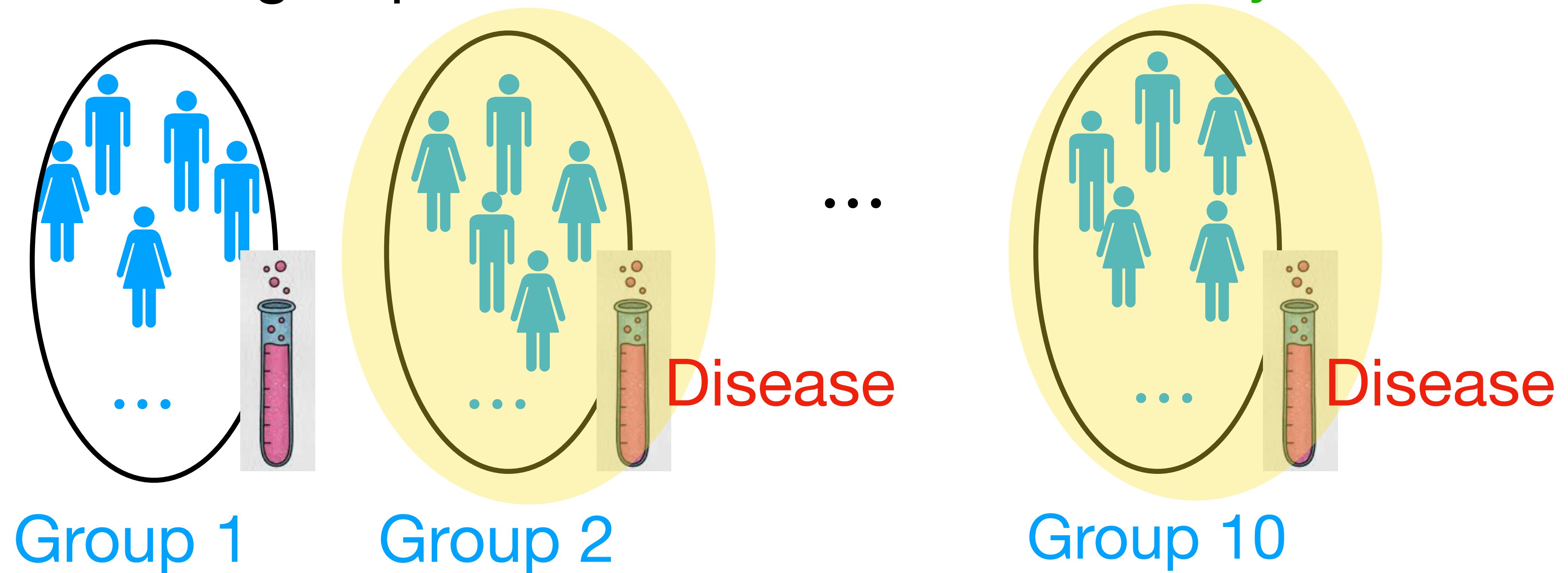


If a group's test shows disease,
test all individuals in group.

WIDELY USED DISCRETE DISTRIBUTIONS: BINOMIAL

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Expected number of tests?

WIDELY USED DISCRETE DISTRIBUTIONS: BINOMIAL

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$X_j = 1$ if j has disease, o.w. $X_j = 0 \Rightarrow$

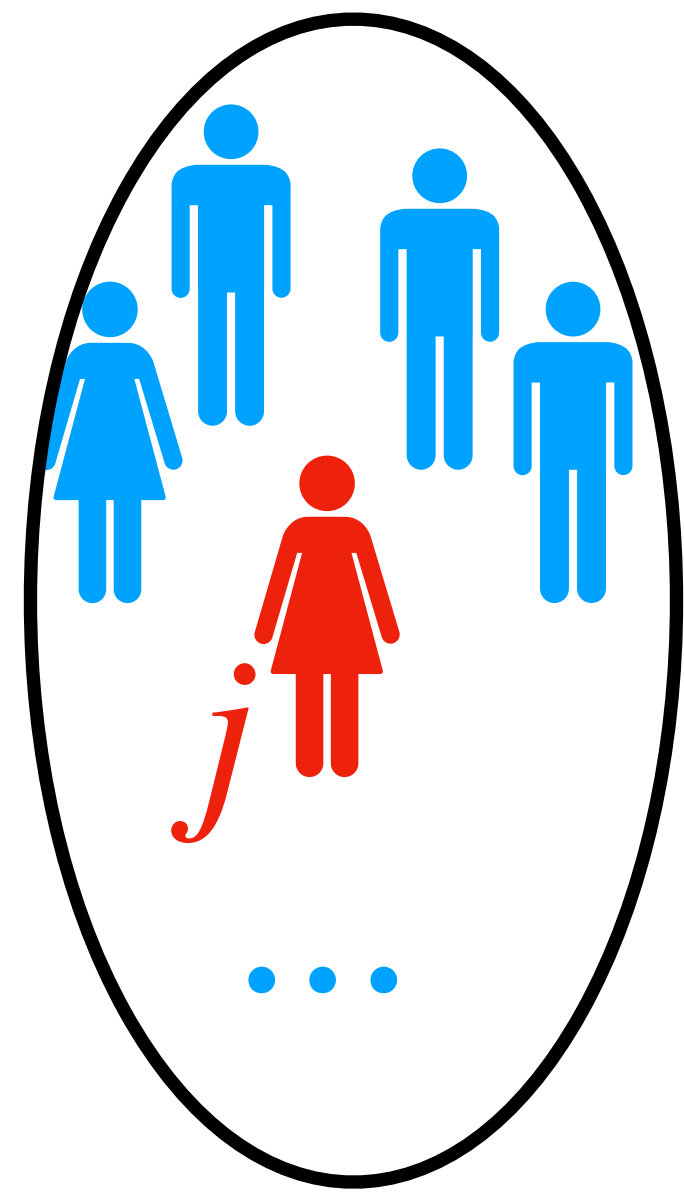
$X_j \sim \text{Bernoulli}(0.002)$, iid for $j = 1, \dots, 1000$



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Group i

$X_j = 1$ if j has disease, o.w. $X_j = 0 \Rightarrow$

$X_j \sim \text{Bernoulli}(0.002)$, iid for $j = 1, \dots, 1000$

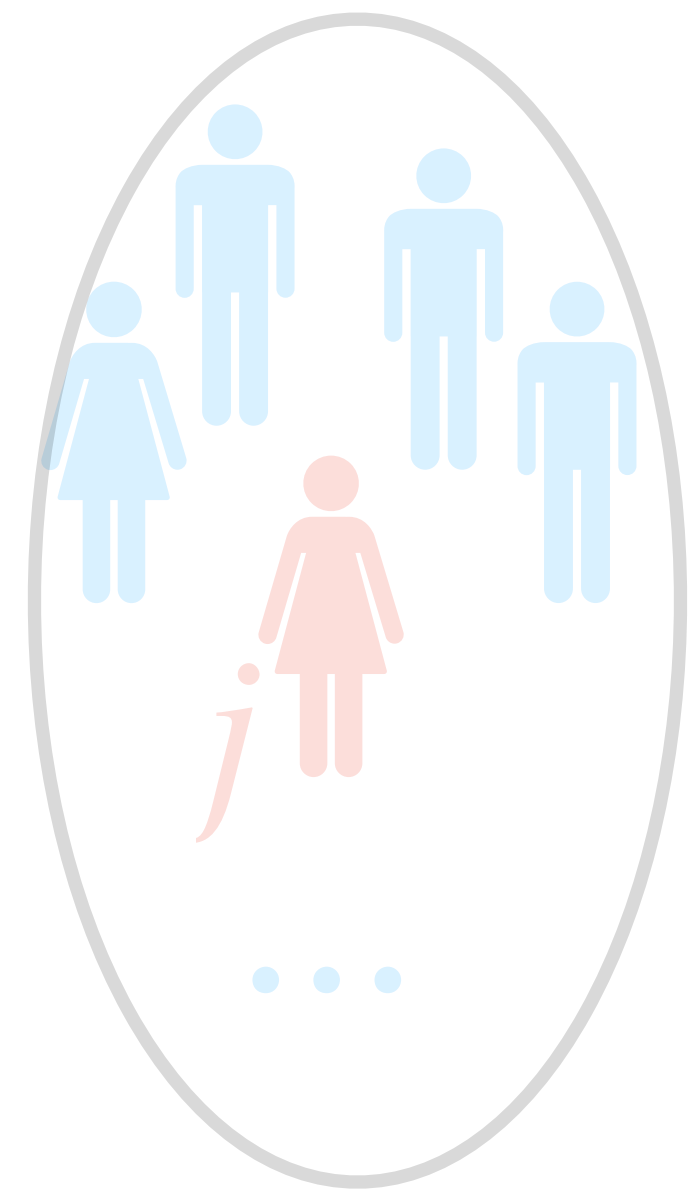
Y_i : # of diseased people in Group $i \Rightarrow$

$Y_i \sim \text{Binomial}(100, 0.002)$ for $i = 1, \dots, 10$

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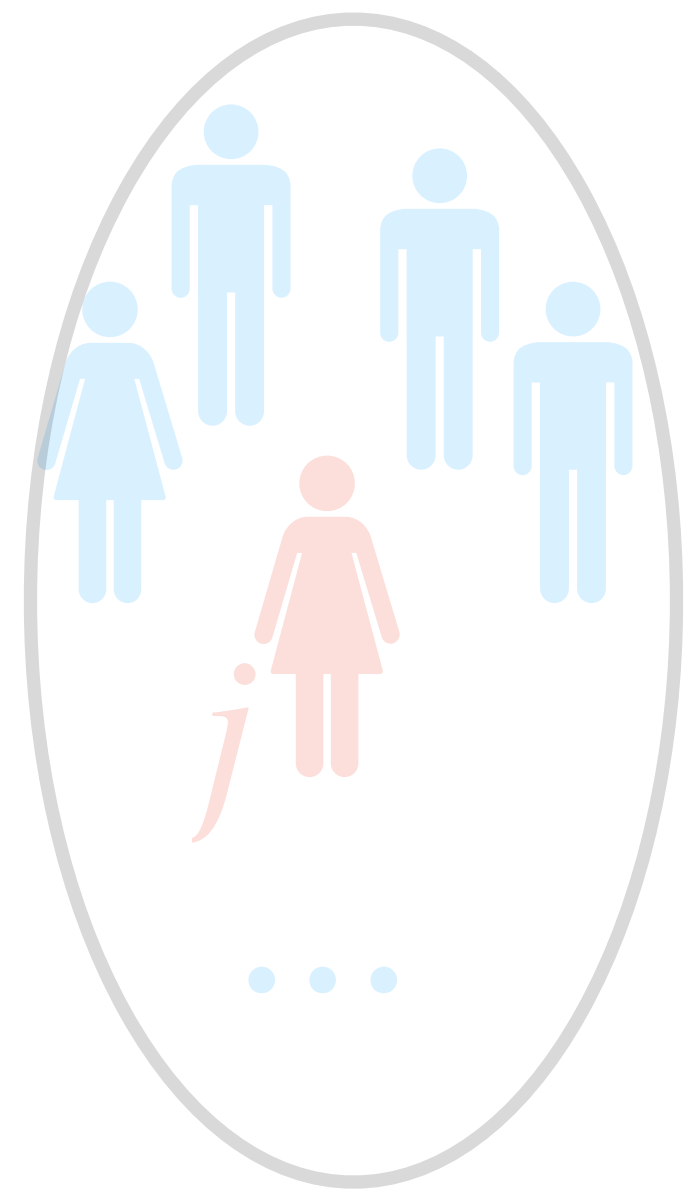
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Y_i : # of diseased people in Group $i \Rightarrow$

$Y_i \sim \text{Binomial}(100, 0.002)$ for $i = 1, \dots, 10$

$Z_i = 1$ if $Y_i > 0$, o.w. $Z_i = 0 \Rightarrow Z_i \sim \text{Bernoulli}(p)$,

where $p = 1 - P(Y_i = 0)$

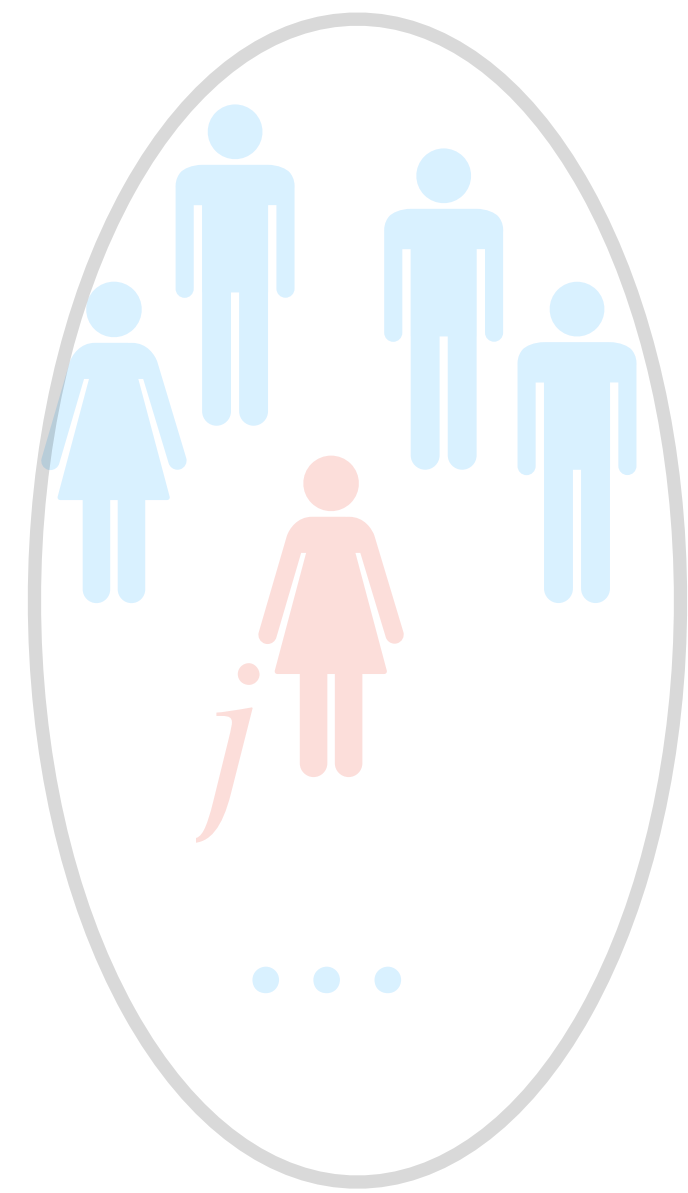
```
from scipy.stats import binom
print(1-binom.pmf(0, 100, 0.002))
```

0.18143319531157243

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Group i

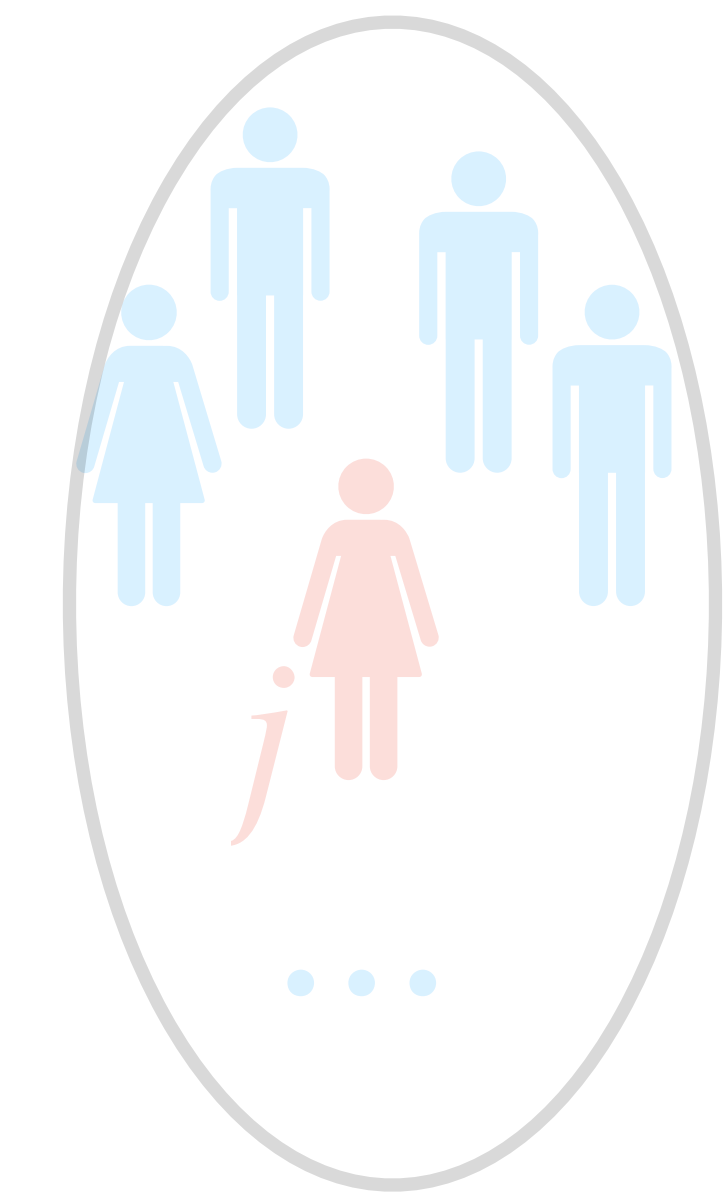
$$Z_i = 1 \text{ if } Y_i > 0, \text{ o.w. } Z_i = 0 \Rightarrow Z_i \sim \text{Bernoulli}(p),$$

$$\text{where } p = 1 - P(Y_i = 0) = 0.181$$

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Naive testing requires 1000 tests. **A better way:**



Group i

$$Z_i = 1 \text{ if } Y_i > 0, \text{ o.w. } Z_i = 0 \Rightarrow$$

$$\text{where } p = 1 - P(Y_i = 0) = 0.181$$

$$W = Z_1 + \cdots + Z_{10} \Rightarrow W \sim \text{Binomial}(10, 0.181)$$

$$E(W) = 10 \times 0.181 = 1.81$$

$$E(100W) = 181, \quad \text{Expected \# of tests} = 191$$